GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 06

Third Term Test (Full Syllabus)

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

Instructions:

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of π as $\frac{22}{7}$. The use of calculators is not allowed.

Part	A	B1	B2	B3	B4	B5	B6	Total
Marks	/40	/10	/10	/10	/10	/10	/10	/100

Part A

Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)

1. Find the square root of 1.2321.

Answer: .....

2. Find the value of: $1\frac{3}{5}$ of $(2\frac{1}{2} - 1\frac{7}{8})$

Answer: .....

3. Expand: $(2a + 3)(a - 4)$

Answer: .....

4. Factorise: $4x^2 - 12x + 9$

Answer: .....

5. Find the least common multiple of the expressions $x^2 + 3x$, $x^2 - 9$ and $2x$.

Answer: .....

6. Simplify: $\frac{x + 1}{x} - \frac{x - 1}{x + 1}$

Answer: .....

7. Find the perimeter of a semicircular plate of radius 7 cm.

Answer: .....

8. In triangle ABC, $A\hat{B}C = 90°$. The magnitude of $B\hat{A}C$ is twice that of $A\hat{C}B$. Find the magnitude of $B\hat{A}C$.

Answer: .....

9. If $\log_2 (x - 3) = 3$, find the value of x.

Answer: .....

10. Find the value of: $\log_2 48 - \log_2 3$

Answer: .....

11. Write the nth term of the arithmetic progression 4, 9, 14, ... in terms of n.

Answer: .....

12. If $\xi = \{a, b, c, d, e, f\}$, $A = \{a, b, c\}$ and $B = \{b, d, e\}$, write $(A \cup B)'$ listing its elements.

Answer: .....

13. An amount of money deposited for 3 years at a simple interest rate of 6% per annum became Rs. 14 160 with the interest. Find the amount deposited.

Answer: .....

14. 15 people started a job that they could complete in 24 days. After 6 days, 3 of them left. Find the number of additional days needed to complete the remaining work.

Answer: .....

15. A vehicle travelled 60 km at a speed of 40 km per hour and the next 60 km at a speed of 60 km per hour. Find the average speed for the whole journey.

Answer: .....

16. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ Make u the subject of the formula.

Answer: .....

17. Find the probability that the sum of the numbers obtained is 7 when two fair dice are thrown together.

Answer: .....

18. In a pie chart showing information about 240 students, the angle at the centre of one sector is 75°. Find the number of students represented by that sector.

Answer: .....

19. The area of triangle ABC is 30 cm^2. Find the area of a parallelogram standing on the base BC and lying between BC and the line parallel to it through A.

Answer: .....

20. In a plan drawn to a scale of 1 : 200, how many centimetres represent the length of a room that is 6 m long?

Answer: .....

Part B

Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)

1. (a) Simplify: $\frac{x^2 - 25}{2x^2 - 9x - 5} \div \frac{x + 5}{2x + 1}$ (3 marks)

(b) A father's present age is 4 times his son's age. After 5 years, the father's age will be 3 times the son's age. Taking the son's present age as s and the father's present age as f, form a pair of simultaneous equations, solve them and find the present ages of both. (4 marks)

(c) Solve: $2x^2 + 5x - 12 = 0$ (3 marks)

2. An incomplete table of values prepared for drawing the graph of the function $y = 4 - \frac{1}{2}x^2$ is given below.

x	−4	−3	−2	−1	0	1	2	3	4
y	−4	−0.5	.....	3.5	4	3.5	2	.....	−4

(a) Complete the blanks in the table. (1 mark)

(b) Draw the graph taking 10 small squares along the x-axis to represent 1 unit and 10 small squares along the y-axis to represent 1 unit. (3 marks)

(c) Using the graph,

(i) write the maximum value of the function and the equation of the axis of symmetry. (1 mark)

(ii) $4 - \frac{1}{2}x^2 = 0$, write the approximate roots of the equation, and from them write an approximate value for $\sqrt{8}$. (2 marks)

(iii) write the range of values of x for which $y > 2$. (1 mark)

(iv) write the value of y when $x = 1.5$. (1 mark)

(d) Write the function of the graph obtained when the graph is moved along the y-axis so that it passes through the origin. (1 mark)

3. A cuboid-shaped wooden block of length 14 cm, width 14 cm and height 10 cm has had a cylindrical hole of radius 3.5 cm drilled through its centre from top to bottom and the wood removed.

(a) Find the volume of the wooden block before drilling. (1 mark)

(b) Find the volume of wood removed. (2 marks)

(c) Find the volume of the remaining piece of wood. (1 mark)

(d) Find the total surface area of the remaining piece of wood, including the inner surface of the hole. (4 marks)

(e) Find the cost of painting that entire surface at Rs. 2 per cm^2. (2 marks)

4. In triangle ABC, D is the midpoint of side BC and E is the midpoint of AD.

(a) Show, with reasons, that the areas of triangles ABD and ADC are equal. (2 marks)

(b) Prove that the area of triangle BEC = $\frac{1}{2}$ × the area of triangle ABC. (3 marks)

(c) If the area of triangle ABC is 64 cm^2, find the area of triangle ABE. (2 marks)

(d) A point F lies on the line drawn through A parallel to BC. Find, with reasons, the area of triangle FBC, and also find the area of triangle FBD. (3 marks)

5. The following table shows information about the number of books read by 40 students last month.

Number of books	0	1	2	3	4	5
Number of students	3	7	12	9	6	3

(a) Write the mode. (1 mark)

(b) Find the median. (2 marks)

(c) Find the mean number of books read by a student. (3 marks)

(d) Find the percentage of students who read more than 3 books. (2 marks)

(e) If every student read 2 more books this month than last month, find the mean for this month. (2 marks)

6. In a pile of logs, the bottom row has 25 logs, and each row going upwards has 2 logs fewer than the row before it. The top row has 1 log.

(a) Find the number of logs in the 6th row from the bottom. (2 marks)

(b) Find the number of rows in the pile. (2 marks)

(c) Find the total number of logs in the pile. (3 marks)

(d) If the upper rows are removed so that only 8 rows remain from the bottom, find the number of logs removed. (3 marks)

*** All rights reserved ***